

MACHINEPULSE · REQUIREMENTS

MachinePulse requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	R4 restated by Amish on 2026-10-02 as a relative energy trend per shift, kWh reported as an estimate; met by design as restated
0.5	2026-10-01	Amish Chadha	Status from MPL-CAL-001 v0.3 for the constructable design (MPL-DDR-003); R16 reported against the value-engineering target; targets unchanged
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Status from MPL-CAL-001 for TRL 3; decisions from MPL-DDR-001 reflected; targets unchanged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-25	Amish Chadha	Initial scaffold

MachinePulse requirements

These are the MachinePulse requirements with their status from the TRL 3 calculation note MPL-CAL-001 v0.3, which checks the constructable design of MPL-DDR-003. The design they are checked against follows the decisions Amish made on 2026-09-25 (MPL-DDR-001 and MPL-DDR-002): Wi-Fi, USB adapter, IIS3DWB-class sensor, one CT, baseline alerts, TwinKit or any broker, stock box with stand-offs and high-temperature magnets. Two targets change in v0.4 under MPL-DDR-002: R8 is restated with a wider tolerance above 85 °C (N2), and R16 follows the budget raised to \$81 for the high-temperature magnets (N1). R3 and R5 keep their targets (N3). One requirement is not met (R10), two are at risk (R3 and R5), nine are met by calculation and four by design. R16 is reported against the value-engineering target: the constructable design is estimated at \$83.50, \$2.50 over the \$81 target. R4 was restated by Amish on 2026-10-02 as a relative energy trend for the first build, with any kWh figure reported as an estimate (MPL-DDR-001, O2), so it moves from at risk to met by design.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Status (MPL-CAL-001)	Verification after TRL 3
R1	Detect run state	Off, idle and running classified from current, with 1 s time resolution	Met by calculation: 1 s RMS windows of 2,000 samples; zero-current reading 0.07 A against a 0.5 A threshold	Bench test on a motor
R2	Log run time	Run hours per machine, error within 1 % over a week	Met by calculation: worst case 1.00 % on 100 s runs, 0.17 % on 10 min runs	Comparison with a timed log
R3	Measure load current	Within 5 % of reading from 10 % to 100 % of CT range, after a two-point calibration	At risk: RSS 3.6 % at 10 % of range and 2.1 % at the design point, but 7.0 % worst case at 10 %; ESP32-S3 ADC residual assumed	Comparison with a clamp meter
R4	Estimate energy per shift	A relative energy trend per shift for the first build, with any kWh figure reported as an estimate (restated by Amish, 2026-10-02, from "within 10 % of a reference meter"; a voltage reference stays an option for sites that need absolute energy)	Met by design as restated: one CT with a load-dependent power factor curve gives the trend; as an absolute figure the estimate is 8.5 % RSS, 16.0 % worst case (26 % with a fixed power factor)	Comparison with a reference meter
R5	Measure vibration	Velocity RMS over 10 to 1,000 Hz (ISO 20816-1 band), three axes	At risk: sensor covers the band, but the magnet mount resonates at about 892 Hz on a painted curved frame (1,783 Hz flat), so readings are 10 % high from about 267 Hz	Mounted shaker check
R6	Low vibration noise floor	0.1 mm/s RMS or better, 10 to 1,000 Hz	Met by calculation: 0.037 mm/s (ADXL345-class fallback 0.20 mm/s would miss it)	Mounted noise test
R7	Resolve running-speed peaks	Spectrum bin 0.5 Hz or finer up to 1 kHz	Met by calculation: 0.407 Hz bins	Spectrum of a known tone
R8	Measure frame temperature	Within 2 °C of the surface from 0 to 85 °C, and within 4 °C from 85 to 100 °C (restated per MPL-DDR-002, N2; was within 2 °C from 0 to 100 °C)	Met by calculation: 1.56 K at 80 °C, 1.69 K at 85 °C and 3.59 K at 100 °C	Comparison with a thermocouple
R9	Report often enough	One summary per minute; spectrum every 15 min; delivered within 2 min	Met by calculation: 1.15 MB per day; 66 s worst delivery	Network log
R10	Install without opening live enclosures	Pod and probe in 10 min without tools; CT on an accessible insulated conductor	Not met for many machines. Pod and probe need no tools; the CT needs a single insulated conductor, often only inside a terminal box or panel opened by a qualified person	Design review with the pilot site
R11	Stay attached	No slip under 5 g peak on a painted cast frame; option for aluminium frames	Met by calculation: slip margin 3.0 on a curved painted frame (2.8 at 80 °C); 3 mm steel pads for aluminium frames, margin 7.0	Pull and shake tests
R12				Hot-plate test

ID	Requirement	Target	Status (MPL-CAL-001)	Verification after TRL 3
	Tolerate hot frames	Operate on frames up to 80 °C in 40 °C ambient	Met by calculation: with 5 mm stand-offs the box floor reaches 53.9 °C and the module about 58.9 °C; the 120 °C magnets (MPL-DDR-002, N1) reach their limit at a 122.7 °C frame, so the accelerometer now sets the limit at 109.6 °C (N-grade magnets would have stopped at 81.3 °C)	
R13	Workshop protection	IP54 or better for the pod	Met by design: IP54 box, three IP68 glands with one cable each, silicone grommet round the sensor boss, sealed light pipe	IP test
R14	Keep data local	Works with TwinKit or any MQTT broker on the local network; no cloud account	Met by design	Configuration check
R15	Store data when the network is down	7 days of summaries on the pod	Met by calculation: 13.9 days of summaries in 4 MB (4.0 days with spectra)	Offline run
R16	Low cost and buildable	Parts \$81 or less per machine (was \$80; MPL-DDR-002, N1); hand tools and a drill press; no custom PCB	Over the value-engineering target by \$2.50: estimated \$83.50 against the \$81 target (indicative prices); hand tools, a drill press and perfboard, no custom PCB	Supplier quotes
R17	Safe by design	Low voltage in the pod; certified 5 V adapter; voltage-output CT only; no connection to machine controls	Met by design; the CT input is now clamped against starting current	Design review

■ Assumptions

- Target machines are driven by induction motors from 0.37 kW to 15 kW, running at 50 or 60 Hz, at speeds from about 700 to 3,600 rpm.
- The CT is chosen per machine from the 5 to 60 A voltage-output range so that normal load sits above about 30 % of its range.
- A single CT on one phase is enough to judge run state and relative load on a three-phase motor whose phases are balanced; it is not an energy meter. Energy uses a power factor curve built from the nameplate and the measured no-load current.
- The pod is mounted on a clean flat or gently curved area of the frame or bearing housing, near the non-drive or drive-end bearing, with paint thickness under about 0.2 mm.
- The workshop has Wi-Fi coverage at the machine, or a TwinKit gateway nearby.